Architecting Reproducible Science: A Practical Path Beyond the Notebook

Software engineering practices for reliable HPC workflows

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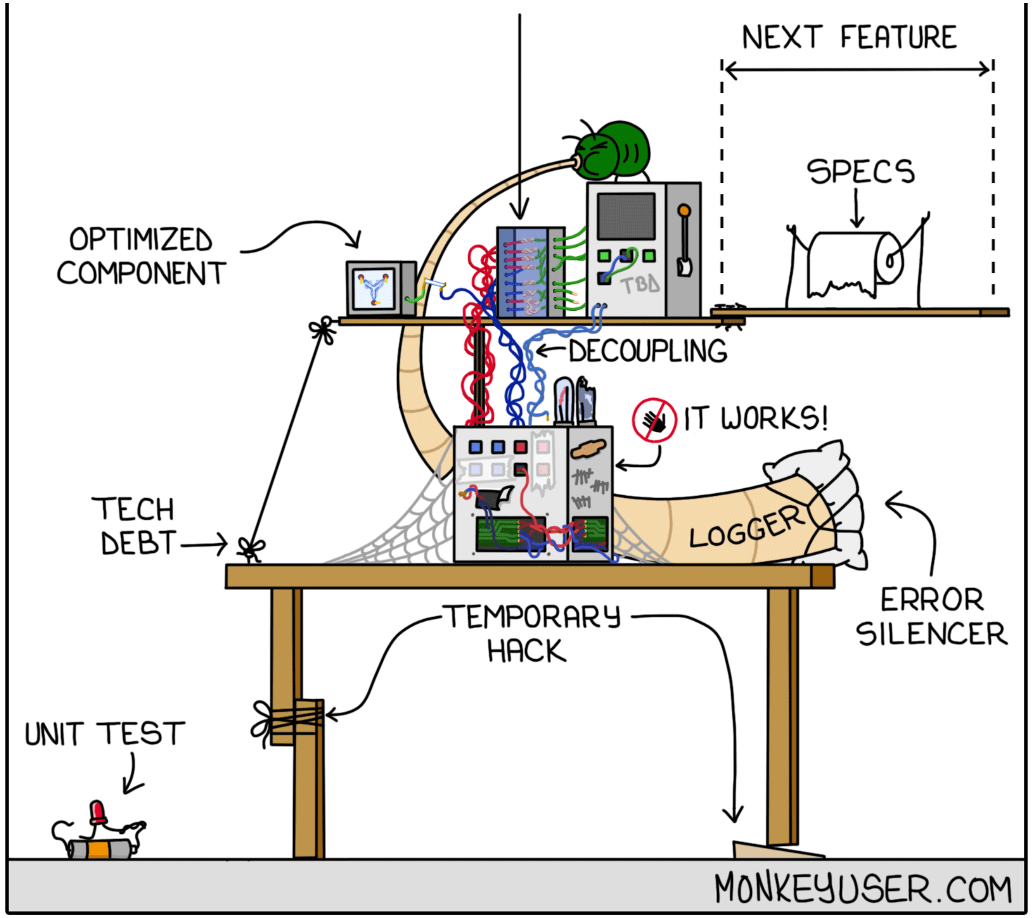
Who is Fernando Garzon?

- Software engineer and computational scientist
- Research Specialist at SDSC
- TSCC user support
- SDSC software development
- DevOps, automation, cloud, and HPC

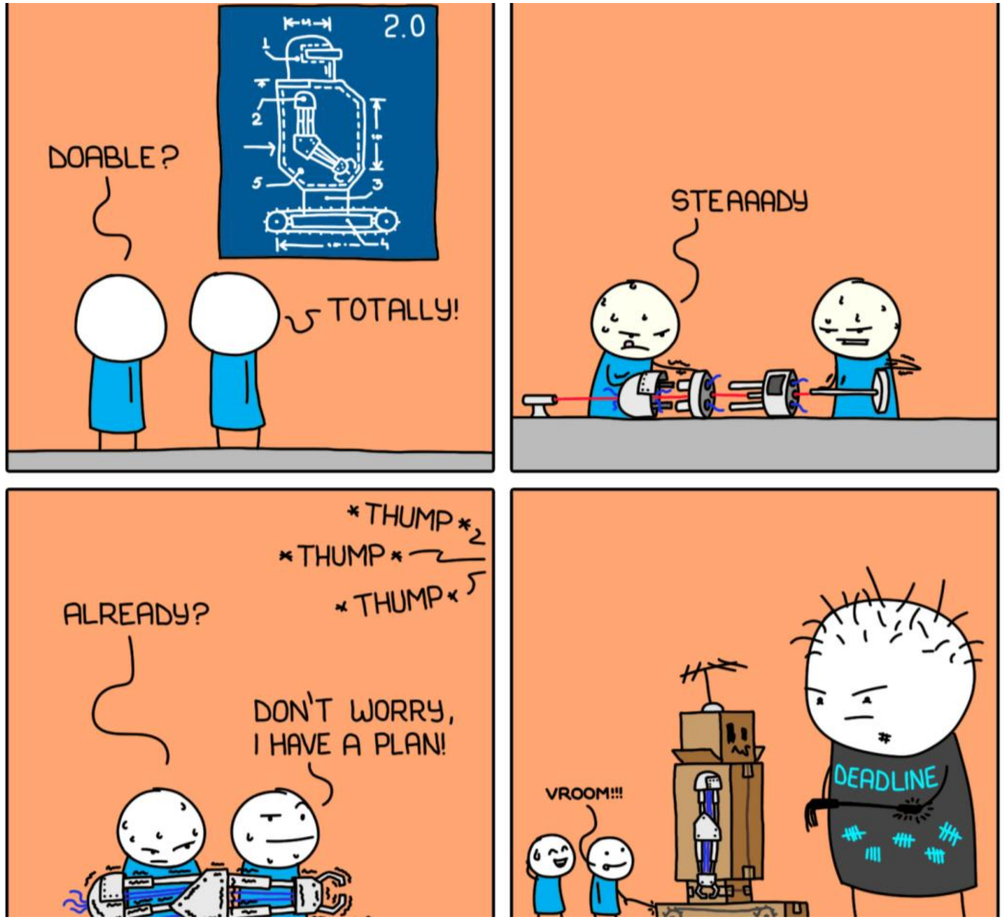
Outline

1. Intro: the reproducibility horror
2. Notebooks and packages
3. Let's Prod This: notebook to package
4. Lights, Camera, GitHub Actions!
5. MLOps: package to container to HPC
6. Final remarks, questions, and tutorial

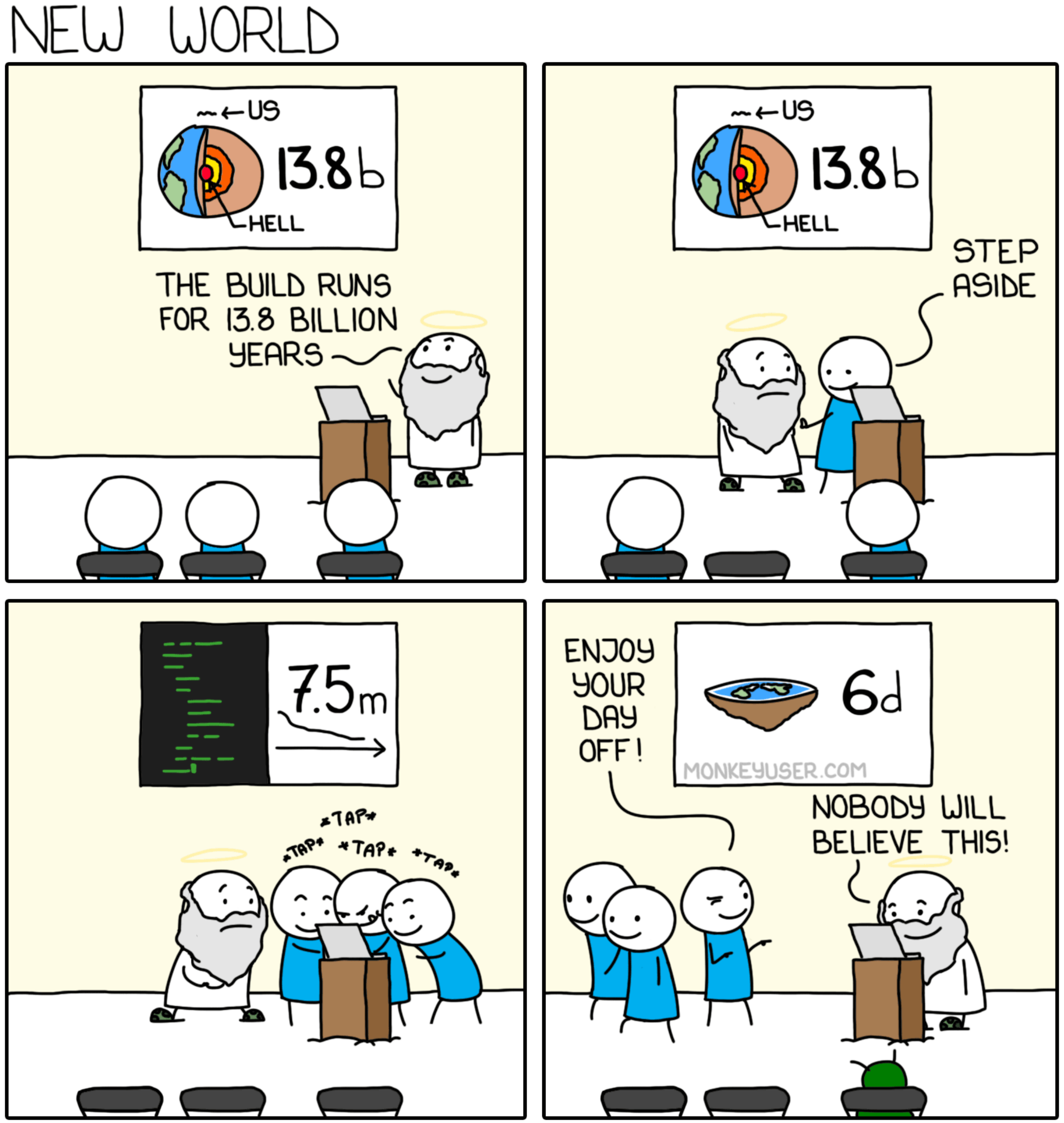
Intro: the reproducibility horror



Notebooks are laboratories. Packages are reliable building blocks.



Let's Prod This



A package turns hidden state into an explicit contract

Lights, Camera, GitHub Actions!

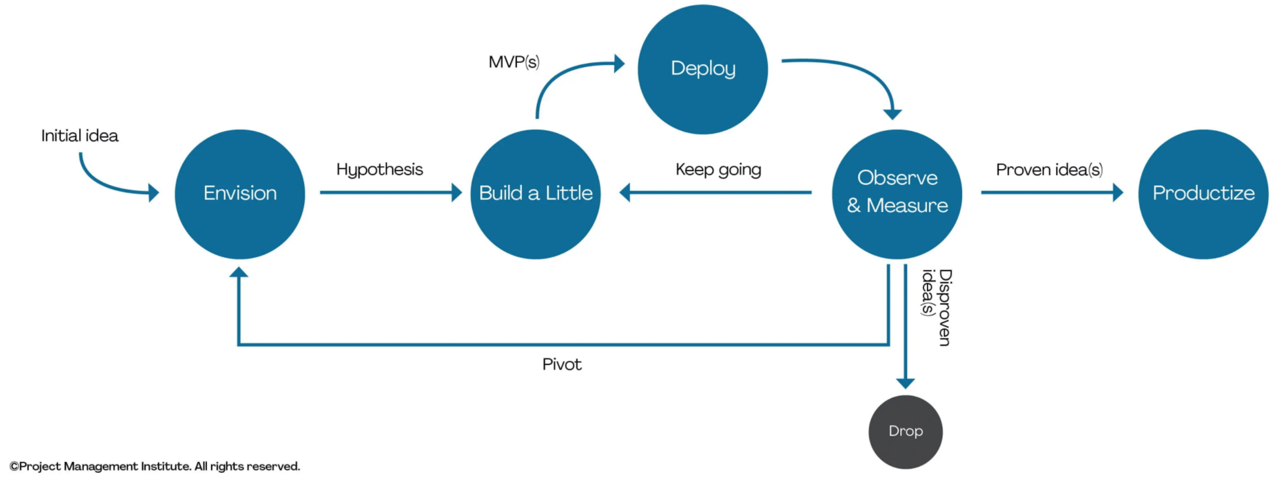
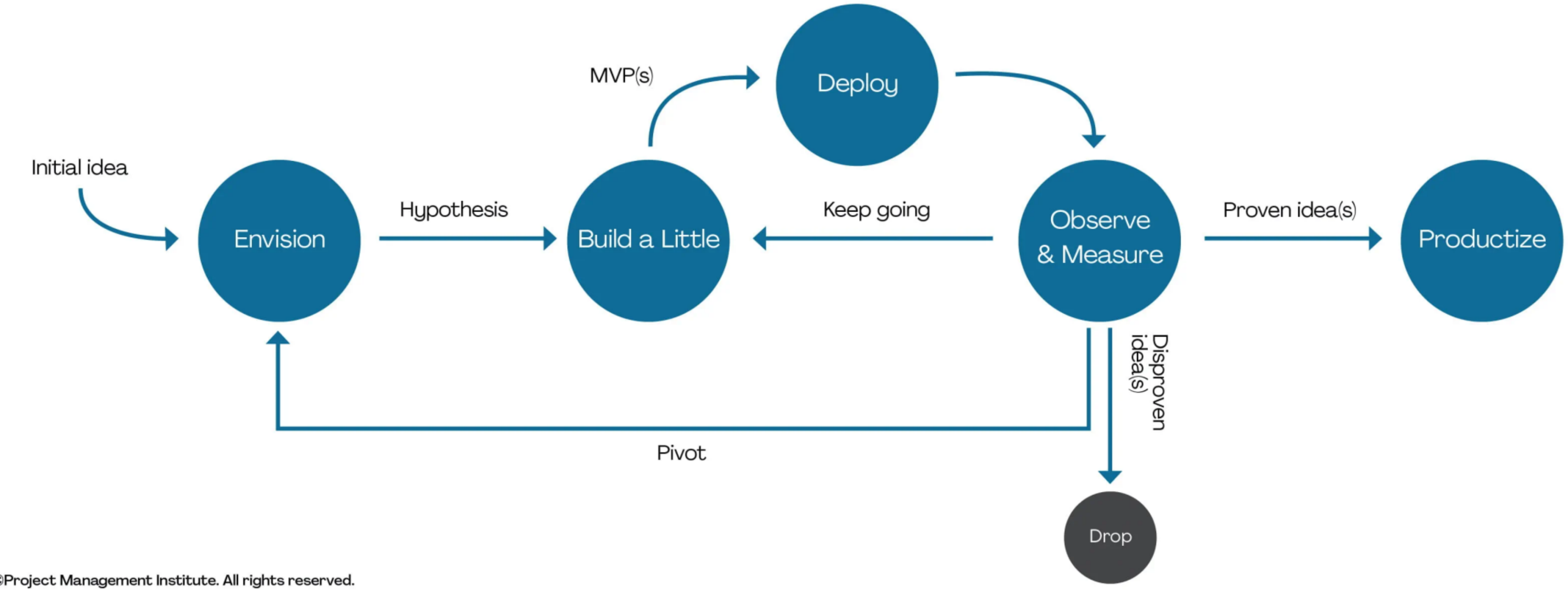


Image reference: https://www.pmi.org/disciplined-agile/lifecycle/exploratory-lifecycle

Tests and CI turn changes into controlled releases

Fail fast. Build once. Release deliberately.



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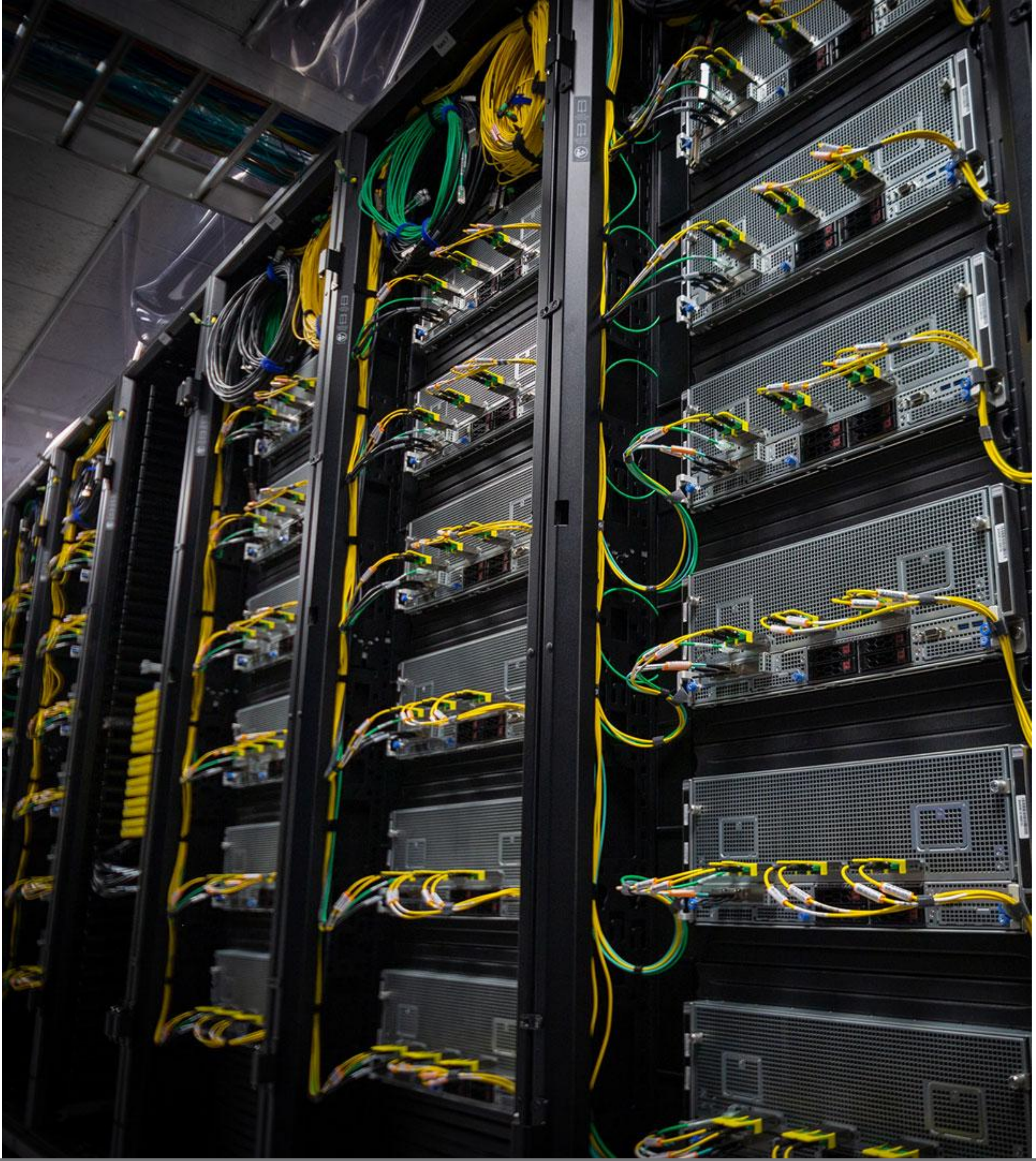
MLOps: one tested artifact, many deployments

1. Package the model.
2. Test the contract.
3. Build one artifact.
4. Deploy predictably.
5. Observe and improve.

One entry point can run locally, in CI, in a container, or on HPC

The model did not need to change. The workflow did.

- Notebooks remain ideal for exploration; packages make execution explicit.
- Tests protect scientific assumptions before expensive runs.
- CI builds trust; containers preserve the runtime; schedulers scale the work.
- Reproducibility is a workflow, not a copy of the notebook.



Questions

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Your turn: make one scientific assumption executable